## DTC P0601:00 [SKYACTIV-D 1.5]

id0102q2705600

| DTC P0601:00 | PCM memory check sum error |
|---|---|
| DETECTION CONDITION | • PCM internal memory check sum error for **5 s**.<br>**Diagnostic support note**<br>• This is a continuous monitor (CCM).<br>• The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle.<br>• FREEZE FRAME DATA/Snapshot data is available.<br>• DTC is stored in the PCM memory. |
| FAIL-SAFE FUNCTION | • Not applicable |
| POSSIBLE CAUSE | • PCM connector or terminals malfunction<br>• PCM malfunction<br>— PCM internal memory malfunction |
| SYSTEM WIRING DIAGRAM | Not applicable |

### Diagnostic Procedure

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION**<br><br>**Note**<br>• Recording can be facilitated using the screen capture function of the PC.<br><br>• Record the FREEZE FRAME DATA/snapshot data on the repair order. | — | Go to the next step. |
| 2 | **VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |
| 3 | **INSPECT PCM CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the PCM connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to the next step. |
| | | No | Go to the next step. |
| 4 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Clear the DTC from the PCM memory using the M-MDS.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Perform the KOEO or KOER self test.<br>(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)<br>• Is the same DTC present? | Yes | Replace the PCM, then go to the next step.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 5 | **VERIFY AFTER REPAIR PROCEDURE**<br>• Perform the “AFTER REPAIR PROCEDURE”.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | DTC troubleshooting completed. |